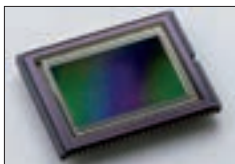


ond, or are looking to win a few awards, opt for the dSLR. One feature that P&S cameras often just don't match up to is the resolution and quality of the photograph taken by a dSLR particularly, if you're keen to blow up the picture to poster size or better. The reason lies deep down in the camera – in the sensor.

14.2 Kinds of Sensors

Digital cameras are basically small computers that convert live images into digital files, by detecting photons (light) which hit the face of an electronic image sensor. As it is with film, the sensor is positioned exactly at the focal point/plane of the camera lens. However, unlike film, the sensor never needs to be moved or changed and is fixed in there permanently.



Sensors make more sense than film

How sensors work

The face of the sensor is made up of millions of tiny 'photosites' – think of them as 'wells' that fill up with light – which are nothing but light-sensitive transistors (semiconductor devices which can amplify or switch electronic signals – and please don't ask us what a semiconductor is). Each individual photosite represents a single pixel and often when manufacturers refer to 'megapixels', they're actually talking about the number of these microscopic photosensitive sensor spots on the face of the sensor.

The sensors themselves are usually quite small – anything from 1/6th to 4/3rd of an inch, typically of the aspect ratio 4:3 (more about aspect ratios later). Even smaller sensors have already been created (such as the Pixelplus 1/10 inch optical format sensor based on 2.2-micron pixel size). The camera creates the image from the array of pixels by scanning each pixel electronically. When the shutter snaps open, a chip inside the camera (the 'image processor') reads the data that the image sensor receives. This little computer then analyses the data from neighbouring pixels to arrive at